

Exp No: 11: STUDY OF GPIO & BIT ADDRESSING IN AT89C51

PROGRAM:

```
#include <reg51.h>
```

```
sbit LED1 = P1^0;
```

```
sbit LED2 = P1^1;
```

```
void delay()
```

```
{
```

```
unsigned int i,j;
```

```
for(i=0;i<200;i++)
```

```
for(j=0;j<1000;j++);
```

```
}
```

```
void main()
```

```
{
```

```
while(1)
```

```
{
```

```
LED1 = 1;
```

```
LED2 = 0;
```

```
delay();
```

```
LED1 = 0;
```

```
LED2 = 1;
```

```
delay();
```

```
}
```

```
}
```

Output :

The image displays two screenshots of the uVision IDE, showing the execution of a C program. The top screenshot shows the program running with a 'Parallel Port 1' dialog box open, displaying the state of the port. The bottom screenshot shows the program running with the same dialog box open, but with different values displayed. Both screenshots show the assembly code and the registers.

Top Screenshot:

- Registers:** R0: 0x00, R1: 0x00, R2: 0x00, R3: 0x00, R4: 0x00, R5: 0x00, R6: 0x00, R7: 0x00, R8: 0x00, R9: 0x00, R10: 0x00, R11: 0x00, R12: 0x00, R13: 0x00, R14: 0x00, R15: 0x00, R16: 0x00, R17: 0x00, R18: 0x00, R19: 0x00, R20: 0x00, R21: 0x00, R22: 0x00, R23: 0x00, R24: 0x00, R25: 0x00, R26: 0x00, R27: 0x00, R28: 0x00, R29: 0x00, R30: 0x00, R31: 0x00.
- Assembly:** C:\0x0000 00 NOP, C:\0x0001 00 NOP, C:\0x0002 00 NOP, C:\0x0003 00 NOP, C:\0x0004 00 NOP, C:\0x0005 00 NOP, ...
- exp.c:** 1 #include <reg51.h>, 2 #bit LED1 = P1^0;, 3 #bit LED2 = P1^1;, 4 void delay(), 5 {, 6 unsigned int i,j;, 7 for(i=0;i<200;i++), 8 for(j=0;j<1000;j++), 9 {, 10 void main(), 11 {, 12 while(1), 13 {, 14 LED1 = 1;, 15 LED2 = 0;, 16 delay(), 17 LED1 = 0;, 18 LED2 = 1;, 19 delay(), 20 }, 21 }
- Parallel Port 1:** Port 1: P1: 0x00, P2: 0x00, P3: 0x00, P4: 0x00, P5: 0x00, P6: 0x00, P7: 0x00, P8: 0x00, P9: 0x00, P10: 0x00, P11: 0x00, P12: 0x00, P13: 0x00, P14: 0x00, P15: 0x00.

Bottom Screenshot:

- Registers:** R0: 0x00, R1: 0x00, R2: 0x00, R3: 0x00, R4: 0x00, R5: 0x00, R6: 0x00, R7: 0x00, R8: 0x00, R9: 0x00, R10: 0x00, R11: 0x00, R12: 0x00, R13: 0x00, R14: 0x00, R15: 0x00, R16: 0x00, R17: 0x00, R18: 0x00, R19: 0x00, R20: 0x00, R21: 0x00, R22: 0x00, R23: 0x00, R24: 0x00, R25: 0x00, R26: 0x00, R27: 0x00, R28: 0x00, R29: 0x00, R30: 0x00, R31: 0x00.
- Assembly:** C:\0x0000 00 NOP, C:\0x0001 00 NOP, C:\0x0002 00 NOP, C:\0x0003 00 NOP, C:\0x0004 00 NOP, C:\0x0005 00 NOP, ...
- exp.c:** 1 #include <reg51.h>, 2 #bit LED1 = P1^0;, 3 #bit LED2 = P1^1;, 4 void delay(), 5 {, 6 unsigned int i,j;, 7 for(i=0;i<200;i++), 8 for(j=0;j<1000;j++), 9 {, 10 void main(), 11 {, 12 while(1), 13 {, 14 LED1 = 1;, 15 LED2 = 0;, 16 delay(), 17 LED1 = 0;, 18 LED2 = 1;, 19 delay(), 20 }, 21 }
- Parallel Port 1:** Port 1: P1: 0x00, P2: 0x00, P3: 0x00, P4: 0x00, P5: 0x00, P6: 0x00, P7: 0x00, P8: 0x00, P9: 0x00, P10: 0x00, P11: 0x00, P12: 0x00, P13: 0x00, P14: 0x00, P15: 0x00.